Continuity and the Intermediate Value Theorem

Section 1: Three-Part Definition of Continuity

1 Which three conditions must be satisfied for a function f to be continuous at $x = c$?

a. $f(c)$ is defined, $\lim_{x \to c} f(x)$ exists, and $\lim_{x \to c} f(x) = f(c)$

b. $f'(c)$ exists, $f(c) > 0$, and $\lim_{x \to c} f(x) = 0$

c. $f(c)$ is continuous on (a, b), $f(c) = 0$, and $f'(c) \neq 0$

d. $\lim_{x \to c^-} f(x) = \lim_{x \to c^+} f(x)$ only

2 Let $f(x) = \frac{x^2 - 9}{x - 3}$ for $x \neq 3$. What value must be assigned to $f(3)$ to make f continuous at $x = 3$?

a. 3

b. 6

c. 0

d. No value can make it continuous

3 The function $g(x) = \frac{1}{x - 2}$ has what type of discontinuity at $x = 2$?

a. Infinite (essential) discontinuity

b. Removable discontinuity

c. Jump discontinuity

d. Oscillating discontinuity

Section 2: Piecewise Functions & Continuity

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4 For what value of k is $f(x) = kx^2 + 1$ for $x \leq 2$, and $f(x) = 3x - 1$ for $x > 2$ continuous at $x = 2$?

a. 1

b. 2

c. 3

d. 0

5 If $g(x) = \frac{\sin(ax)}{x}$ for $x \neq 0$, and $g(0) = 4$, what value of a ensures continuity at $x = 0$?

a. 1

b. 2

c. 4

d. $\frac{1}{4}$

6 The function $h(x) = x + 2$ for $x < 1$, and $h(x) = 2x + 1$ for $x \geq 1$ has what property at $x = 1$?

a. Continuous because both one-sided limits equal 3

b. Jump discontinuity of size 1

c. Infinite discontinuity

d. Undefined at $x = 1$

Section 3: Intermediate Value Theorem (IVT)

7 What are the hypotheses of the Intermediate Value Theorem for a function f on interval $[a, b]$?

a. f must be continuous on the closed interval $[a, b]$

b. f must be differentiable on (a, b)

c. $f(a)$ and $f(b)$ must have opposite signs

d. f must be strictly increasing on $[a, b]$

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8 Let f be continuous on $[1, 5]$ with $f(1) = -3$ and $f(5) = 7$. Does the IVT guarantee a solution to $f(c) = 0$ for some $c \in (1, 5)$?

a. Yes, because 0 lies between -3 and 7

b. No, f must also be differentiable

c. No, f must be a polynomial

d. Only if $f'(x) > 0$ for all x

9 On which interval is the polynomial $P(x) = x^3 - 3x - 1$ guaranteed to have a real zero by the IVT?

a. $[1, 2]$

b. $[0, 1]$

c. $[-1, 0]$

d. $[2, 3]$

Section 4: Extreme Value Theorem (EVT)

10 What does the Extreme Value Theorem guarantee for a function f continuous on a closed interval $[a, b]$?

a. f attains both an absolute maximum and an absolute minimum on $[a, b]$

b. f has a horizontal tangent on (a, b)

c. $f'(c) = 0$ for some $c \in (a, b)$

d. f is invertible on $[a, b]$

11 Why does the function $f(x) = \frac{1}{x}$ NOT attain an absolute maximum on $(0, 2]$?

a. The interval is open at 0 and f increases without bound as $x \to 0^+$

b. f is not differentiable on $(0, 2]$

c. f is bounded on $(0, 2]$

d. $f(2) = 0.5$ is the maximum

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12 Find the absolute maximum value of $f(x) = -x^2 + 4x + 1$ on the closed interval $[0, 3]$.

a. 5

b. 4

c. 1

d. 6

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